

	HIGH-END								
Company	ASUS	ASUS	Zotac	Galaxy	XFX	ASUS	ASUS	Zotac	MSI
Model Name	Direct CU II TOP GTX 680	HD 7970 DCII	GTX 680	GTX 680	R7970	Direct CU II TOP GTX 670	HD 7970	GTX 670	R7950 Twin Frozr
Price	40500	38000	35500	35500	48000	32000	37000	29999	41500
Features	1.7	1.9	0.65	0.35	0.4	1.5	0.6	0.35	1.05
Performance	61.19	60.46	60.41	58.68	58.40	58.25	56.12	54.83	53.08
Warranty	3	3	5	NA	2	3	3	5	3
Features									
Chipset	GK104	Tahiti XT	GK104	GK104	Tahiti	GK104	Tahiti XT	GK104	Tahiti Pro
Engine/Memory Clock	1137 / 1502GDDR5	1000 / 1400GDDR5	1006 / 1502GDDR5	1006 / 1502GDDR5	925 / 1375GDDR5	1059 / 1502GDDR5	925 / 1375GDDR5	915 / 1502GDDR5	880 / 1250GDDR5
Stream processors/ Shaders	1536	3096	1536	1536	2048	1344	3096	1344	1792
Texture Units	128	128	128	128	128	112	128	112	112
ROPs	32	32	32	32	32	32	32	32	32
Memory (MB)	4096	3072	2048	2048	3072	2048	3072	2048	3072
Ports - VGA/DVI/HDMI/Mini-HDMI/DP/Mini-DP	N/Yx2/Y/N/Y/N	N/Yx2/N/N/Yx4/N	N/Yx2/Y/N/Y/N	N/Yx2/Y/N/Y/N	N/Y/Y/Yx2/N/N	N/Yx2/Y/N/Y/N	N/Y/Y/N/N/Yx2	N/Yx2/Y/N/Y/N	N/Y/Y/Yx2/N/N
BIOS switches	Y	Y	N	N	N	Y	Y	N	Y
External Power	8PIN+6PIN	2x8PIN	2x6PIN	2x6PIN	8PIN+6PIN	2x6PIN	8PIN+6PIN	2x6PIN	2x6PIN
Adapters - VGA-DVI/HDMI-DVI/MiniDP-DVI	N	N/Y/N/N	Y/N/N/N/N	Y/N/N/N	N/Y/N/N/N	N	N/Y/N/N/Y	Y/N/N/N/N	Y/N/N/N/N
Other accessories	2X6PIN-8PIN PCIe	2x6PIN-8PIN, 2xMolex-6PIN	2XMolex-PIN PCIe	2XMolex-6PIN PCIe	N	2xMolex-6PIN PCIe	2x6PIN-8PIN, 2xMolex-6PIN	2XMolex-6PIN PCIe	Power cable
Temperature (degrees °C)									
Idle / Load	31 / 58	31 / 60	35 / 81	37 / 80	31 / 73	35 / 59	36 / 78	33 / 79	35 / 75
3D Mark 11									
Performance - Score	10873	8695	9551	9541	8319	9421	8334	8528	7651
Graphics	10878	8279	9341	9346	7863	9129	7879	8156	7145
Physics	13029	12916	13062	13023	12960	13033	12941	13014	12969
Combined	8688	7812	7741	7668	7554	8016	7573	7263	7062
Unigine 2.5									
1080p Max FPS	46.9	51.5	52.3	44.2	48.6	44.1	38.6	40.3	35.5
Score	1180	1297	1318	1112	1225	1111	973	1014	895
720p Max FPS	76.1	68.3	76.6	70.7	63.7	70.9	56.3	63.8	52.4
Score	1916	1720	1931	1781	1603	1786	1418	1607	1314
Resident Evil 5 (FPS)									
1080p Max / Med	178.5 / 206.1	177.5 / 216	172.4 / 201.3	169.3 / 204.8	170.4 / 217.6	168.1 / 208.5	171.1 / 216	162.9 / 204.4	160.8 / 217.3
720p Max / Med	178.4 / 198.3	192.5 / 215.4	181.8 / 196.3	177.6 / 200	196.1 / 217.1	176.8 / 198.8	191.7 / 218.7	176.9 / 198.7	194.8 / 214.8
Dirt 3 (FPS)									
1080p Max / Med	110.54/224.1	93.6/189.73	101.84/221.9	102.11/222.4	89.29/183.3	101.5/215.8	89.26/177.3	92.39/181.96	81.58/170.0
720p Max / Med	155.9 / 223.92	134.06 / 189.04	148.32 / 222.18	146.71 / 223.52	128.34 / 211.28	151.4/226.7	127.5 / 210.73	136.99 / 208.33	119.47 / 207.78
Lost Planet 2 (FPS)									
1080p Max / Med / Low	69 / 99.1 / 122.8	60.3 / 87.4 / 113.8	74.8 / 93.6 / 124.6	75.4 / 92.9 / 122.1	57.4 / 84 / 110.9	65.2 / 93.3 / 117.9	57.7 / 84.4 / 113.3	67.6 / 86.7 / 118.6	52.3 / 76.9 / 110.4
720p Max / Med / Low	92.9 / 112.7 / 131.6	77.3 / 99.4 / 133.1	96.4 / 108.9 / 131.5	96.1 / 109.4 / 134.4	73.3 / 97.7 / 121.8	88.5 / 109.1 / 129.7	73.8 / 96.9 / 121.4	88.9 / 104.9 / 135	68.1 / 94.1 / 121.7
Sniper Elite V2 (FPS)									
1080p Max/Med/Low	24.3 / 80.1 / 154.2	48.3 / 88.1 / 177.5	22.6 / 75 / 153.6	22.5 / 75.3 / 151.3	28.8 / 86.8 / 181	21.7 / 74.3 / 151.3	25.2 / 73.3 / 155.1	20.5 / 69.7 / 142.9	25.7 / 78.2 / 161.5
720p Max/Med/Low	47.4 / 144.2 / 244.1	56.8 / 159.2 / 259.5	44.5 / 138.2 / 228.1	44.4 / 140.1 / 226.9	53.8 / 153.6 / 278.9	43.2 / 135.8 / 227.8	46.9 / 134 / 243	40 / 130.2 / 216.2	48.2 / 142 / 261.5
Batman: AC (FPS)									
1080p Max / Med / Low	57/84/115	67/89/109	51/80/99	51/80/98	65/85/165	57/84/106	63/86/107	50/77/95	57/78/97
720p Max / Med / Low	96 / 111 / 123	109 / 117 / 132	91 / 106 / 122	91 / 106 / 121	102 / 108 / 132	97 / 110 / 124	106 / 114 / 121	92 / 105 / 118	97 / 113 / 122
OpenCL/GL Tests									
TessMark (Score)	6488	3666	5976	5984	3399	5103	3396	4631	3222
LuxMark (Samples/sec)	670	1782	618	618	1711	616	1708	563	1541
FurMark (Score)	2692	2585	2667	2657	2583	2325	3349	2323	2101
AESDecrypt (sec)	1.41037	0.988778	1.18052	1.2312	0.780069	1.26116	0.817297	1.22644	0.872272
AESDecrypt (sec)	1.29874	0.982334	1.52814	1.27633	0.787801	1.36917	0.910495	1.37357	0.882075